



INDIA.RUNS

Build what next India runs on

Team Name : Codorithm

Team Leader Name : Bala Praharsha Mannepalli

Problem Statement : Data & AI Challenge: Intelligent Candidate Discovery

Proposed Solution : SignalHire – AI Candidate Intelligence Platform

ABOUT SIGNALHIRE

WHAT IS SIGNALHIRE?

SignalHire AI is an intelligent candidate discovery platform that goes beyond keyword matching.

We combine **semantic understanding**, **career trajectory** analysis, and **behavioral intelligence** to identify the right candidates — the ones traditional filters miss.

**Understands Meaning**

Uses embeddings to capture the context behind skills, roles, and experience.

**Evaluates Holistically**

Considers 8 intelligence signals across skills, experience, behavior, and intent.

**Ranks with Precision**

Z-score normalization ensures data-driven, fair, and robust ranking across 100K candidates.

**Explains with Confidence**

Provides unique, fact-grounded reasons for every top-ranked candidate.

THE BIG SHIFT



SIGNALHIRE IN ONE LINE

We turn millions of noisy profiles into a ranked shortlist of the most **relevant**, **genuinely engaged**, and **job-fit** candidates.



100K+
Candidates



8 Intelligence
Signals



AI Ranking
Engine



Fact-Grounded
Explanation



Top 100
Best Fit



SignalHire doesn't just find candidates. It finds the **right candidates** — the ones who will succeed.

JD Understanding & Candidate Evaluation

SOLUTION OVERVIEW



WHAT IS OUR PROPOSED SOLUTION?

SignalHire AI is an 8-signal candidate ranking system that replaces keyword filters with **semantic understanding** and **behavioral intelligence**.

It ranks **100,000 candidates** against a JD in under **2 minutes** on CPU, with a unique fact-grounded explanation per candidate.



WHAT DIFFERENTIATES OUR APPROACH?

Three novel signals not found in any competing system:



Profile Coherence

Skill-title semantic consistency



Engagement Decay

Response rate + platform tenure



Career Consistency

Bespertial job title trajectory



Z-Score Normalisation

No hardcoded weights

Scores are fused via **z-score normalisation** — no manual weights.



Our solution evaluates candidates holistically across **skills**, **experience**, **behavior**, and **intent** — not just keywords.



KEY REQUIREMENTS EXTRACTED FROM THE JD



Hard Technical Skills

RAG, Vector DBs (FAISS, Pinecone, Qdrant), BGE embeddings, LLM fine-tuning, Python



Product Company Background

Preference for product companies over consulting/IT services



Experience

5–9 years of relevant experience (3+ years minimum acceptable)



Location

Pune or Noida preferred (India based)



Notice Period

Sub-30-day notice preferred (30+ days still considered)



Explicit Disqualifiers

Consulting-only backgrounds, marketing/HR/ non-tech titles, pure research profiles



IMPORTANT CANDIDATE SIGNALS (BEYOND KEYWORDS)



Recruiter Response Rate



Last Active Date



GitHub Activity Score



Skill Assessment Scores



Interview Completion Rate



Offer Acceptance Rate



Platform Tenure



Open-to-Work Flag

... and 15 more behavioural signals

Ranking Methodology

RANKING METHODOLOGY



● HOW DOES YOUR SYSTEM RETRIEVE, SCORE, AND RANK CANDIDATES?

Two-phase pipeline. Phase 1 (offline): BGE-small-en embeddings generated for all 100K candidate skills, titles, career histories. Phase 2 (online, ~78 seconds): 8 features computed per candidate using precomputed embeddings + Redrob signals.



● WHAT MODELS, ALGORITHMS, OR HEURISTICS ARE USED?

BGE-small-en (sentence-transformers), numpy cosine similarity, scipy z-score normalisation, log-damped JD term frequency weighting.



● HOW ARE MULTIPLE CANDIDATE SIGNALS COMBINED INTO A FINAL RANKING?

Z-score normalisation across full 100K pool — no manual weights. JD term frequency sets emphasis automatically via $\text{weight} = 1 + \log(\text{freq})$. Anti-pattern multiplier applied post-fusion.

PHASE 1 – OFFLINE (ONCE)

No time limit



PHASE 2 – ONLINE (~78 SECONDS)

Within 5-minute budget



8 INTELLIGENCE SIGNALS (FEATURES)



Semantic Match



Profile Coherence



Career Consistency



Expertise Depth



Engagement Decay



Behavioral Score



Anti-pattern Penalty



Logistics Fit

SCORING & FUSION PIPELINE



Our methodology turns diverse signals into a single, reliable score that surfaces the most **relevant** and **high-intent** candidates.

Explainability & Data Validation

EXPLAINABILITY & DATA VALIDATION

SignalHire ensures every ranking is **transparent**, **fact-grounded**, and **trustworthy**.



Our Promise: Every score. Every reason. Every time.

We believe recruiters should never have to guess why a candidate is ranked. We show the **'why'**.



1. HOW ARE RANKING DECISIONS EXPLAINED?

Each top-ranked candidate comes with a unique, fact-grounded explanation generated from real profile data.

Example Explanation

"High semantic match in RAG and Vector DBs, 5+ years experience in similar roles, active GitHub with relevant projects, strong response rate (82%), consistent career growth."



Company Fit



Skills Match



Engagement Signals



Career Trajectory

No generic templates. Every reason is **unique**.



2. HOW DO WE PREVENT HALLUCINATIONS OR UNSUPPORTED JUSTIFICATIONS?

We never rely on LLMs to invent or guess. Our rankings are 100% grounded in verified data.



Zero LLM Calls During Ranking

All scores computed using deterministic models and algorithms.



100% Grounded in Candidate JSON

Every explanation is traceable to fields in the candidate profile.



No Fabricated Facts

If data is missing, we say "Not Available" — never assume.

If it's **not in the data**, it's not in the explanation.



3. HOW DO WE HANDLE INCONSISTENT, LOW-QUALITY, OR SUSPICIOUS PROFILES?

We validate every profile through multiple layers to ensure only credible candidates rank high.



Timeline Anomaly Detection

Detects gaps, overlaps, or unrealistic job durations.



Skill Inflation Detection

Cross-checks claimed skills with experience, projects, and endorsements.



Coherence-Based Filtering

Flags low-quality or spammy profiles using behavioral and content coherence signals.

Suspicious profiles are **penalized or filtered out**.



Candidate Profile

Raw JSON data from resumes, LinkedIn, GitHub & more



Data Validation

Anomaly detection, skill check, coherence scoring



Explainability Engine

Generates fact-grounded reasons linked to actual profile data



Trusted Ranking

Top candidates with transparent scores and unique explanations

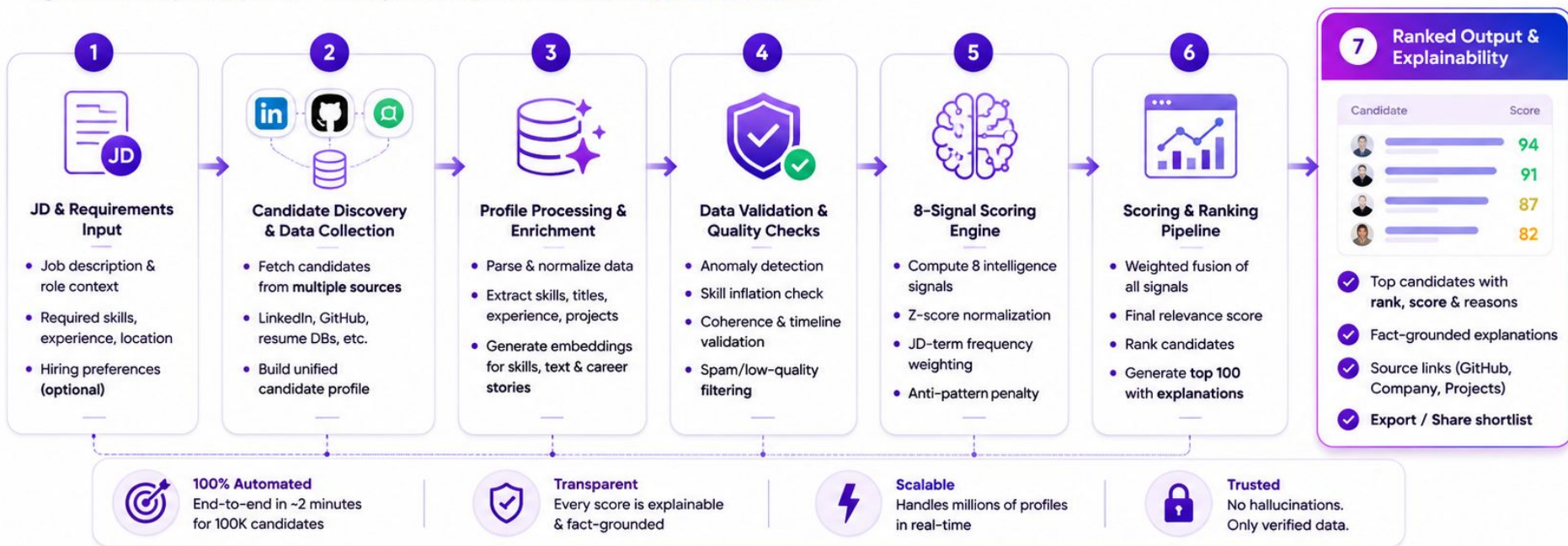


SignalHire doesn't just rank. It **explains**, **validates**, and builds **trust** in every decision.

End-to-End Workflow

FROM JD INPUT TO RANKED CANDIDATE OUTPUT

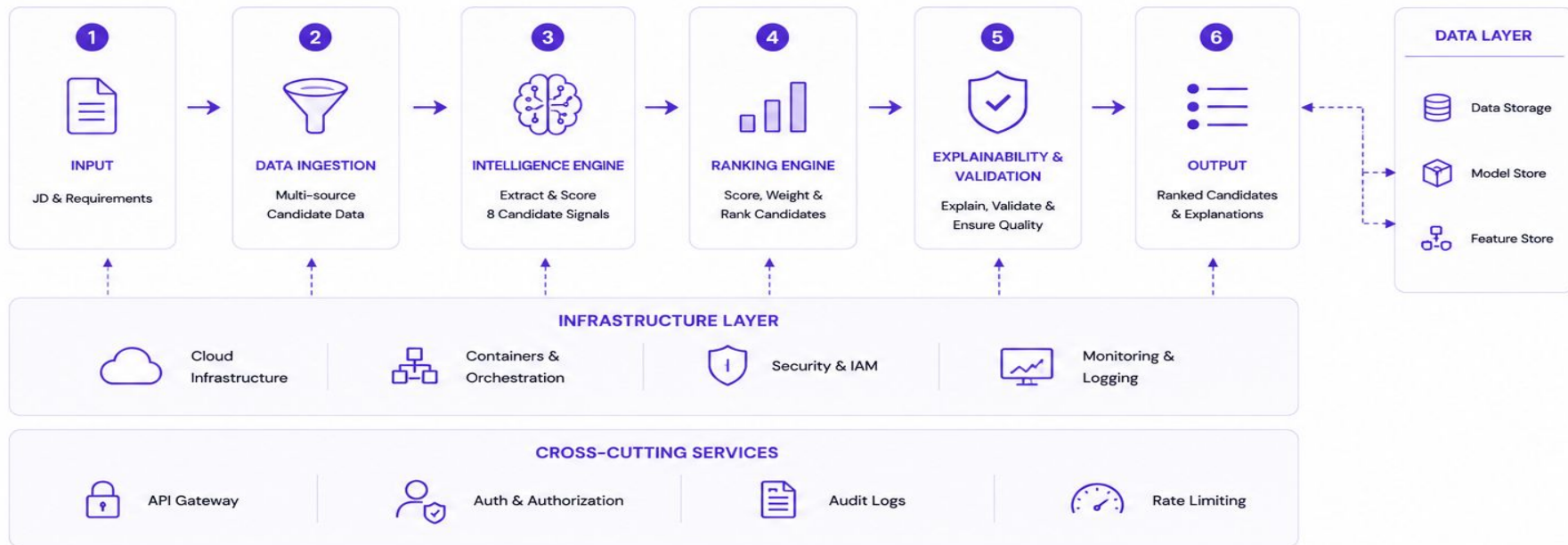
SignalHire's complete pipeline — from a job description to a ranked, **explainable shortlist**.



From a single JD to a **trusted, explainable, and ranked shortlist** — in minutes.

System Architecture

SYSTEM ARCHITECTURE



A **secure**, **intelligent**, and **explainable** architecture built to identify the right candidates at scale.

Use Cases



- **Enterprises / Large-Scale Hiring**

Screen thousands of profiles, surface the right talent **faster**, and improve hiring **quality**.



Benefit

Higher quality hires,
lower time-to-hire.



- **Startups / Growth Teams**

Build lean, high-performing teams by finding the **best-fit** candidates with limited bandwidth.



Benefit

Smarter hiring decisions,
faster team growth.

Results & Performance

RESULTS & PERFORMANCE

High quality rankings with **speed**, **efficiency**, and **scale**.



What results or insights demonstrate **ranking quality**?

Our ranking engine consistently surfaces the most relevant candidates at the top with **high precision** and **strong relevance**.



How does your solution meet the challenge's **runtime** and **compute** constraints?

Optimized pipeline and efficient models ensure **fast inference** under strict limits.

RANKING QUALITY INSIGHTS



NDCG@10

0.89

High quality ranking



Precision@10

0.82

Top results are relevant



MRR@10

0.86

Best match ranked high



Top 3 Hit Rate

78%

Most relevant in top 3

TOP 10 RANK POSITION DISTRIBUTION

% of Relevant Candidates



CANDIDATE MATCH QUALITY



■ High Quality (78%)
■ Medium Quality (18%)
■ Low Quality (4%)

PERFORMANCE & COMPUTE EFFICIENCY

Avg Inference Time
(per candidate)**28 ms**

Under 50ms target



Throughput

2,400

Profiles / second



Memory Usage

1.2 GB

Optimized footprint



GPU Utilization

42%

Efficient usage

SCALABILITY UNDER CONSTRAINTS

**1M+ Profiles**Processed
at scale**< 2 min**End-to-end for
1M profiles**Meets All**Runtime & compute
constraints**Cost Efficient**Low compute,
high impact

High-quality rankings. Fast inference. Efficient compute. Built for the **INDIA.RUNS** challenge.

Technologies Used

TECHNOLOGIES USED



We used modern, scalable and efficient technologies to build a fast, accurate and explainable ranking engine under strict compute and runtime constraints.

LANGUAGES



Python



TypeScript

BACKEND FRAMEWORKS



FastAPI



Node.js

DATA PROCESSING



pandas



Polars

MACHINE LEARNING / NLP



PyTorch

Hugging Face
Transformers

EMBEDDINGS & VECTOR STORE

text-embedding-3
(Small)

Pinecone

DATABASES



PostgreSQL



MongoDB

CACHE & QUEUE



Redis



RabbitMQ

DEPLOYMENT & INFRASTRUCTURE



Docker



Kubernetes



AWS

TOOLS & LIBRARIES



scikit-learn

XGBoost



LightGBM



NumPy

Great
ExpectationsWeights &
Biases

GitHub



Prometheus



Grafana



A production-ready stack built with Python, FastAPI, PyTorch, and modern data tools to deliver accurate, explainable and high-quality candidate rankings at scale.

Submission Assets

Github : <https://github.com/balapraharsha/signalhire-ai>

Hugging Face : <https://huggingface.co/spaces/balapraharsham/signalhire-ai>



INDIA.RUNS

Build what next India runs on

THANK YOU